CORRELATIONS BETWEEN PRIMITIVE ROOT STATUSES OF CONSECUTIVE PRIMES

Abstract. Call a prime p an *Artin prime for base* 10 if 10 is a primitive root modulo p. We investigate the dependence between the Artin statuses of *consecutive* primes. Computing Artin status for all 50,847,531 primes $7 \le p \le 10^9$, we find that consecutive primes are measurably *anticorrelated*: $P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) = 0.36514$ versus $P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n) = 0.37928$, a deficit $\delta = -0.01414$ (Pearson statistic 10,167 on one degree of freedom; signed square root -100.8 under a nominal independent-status reference model; over 50,847,530 consecutive pairs). The dependence has striking gap structure. We prove an elementary exclusion law: if $p, p+g > 5$ are both prime with $g \equiv 20 \pmod{40}$, then 10 is a quadratic residue modulo exactly one of them, so they can never both be Artin primes for base 10; empirically, among 2,195,882 consecutive pairs with gap 20 or 60 there is indeed not a single doubly-Artin pair. Conversely, gaps $g \equiv 0 \pmod{40}$ force equal quadratic-residue status and exhibit strong positive dependence ($P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) = 0.899$ at $g = 40$). Conditioning on the joint residues of (p_n, p_{n+1}) modulo 120 or 840 substantially reduces a selected-cell signed residual association; the remaining association is small in that metric but is not established to vanish. These results are consistent with a substantial role for arithmetic residue structure in the observed anticorrelation; they do not constitute a complete decomposition or an independent quantitative verification of the Hardy–Littlewood explanation. As an intermediate link we report the (apparently unreported) anticorrelation $r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big) = -0.041$. We formulate open questions and discuss extensions to general bases.

1. Introduction

Two classical themes in the theory of primes intersect in this paper.

The first is Artin's primitive root conjecture, communicated to Hasse in 1927 (see [1] and the historical account in [14]): for any integer $a \notin \{-1, 0, 1\}$ that is not a perfect square, the set of primes p for which a is a primitive root modulo p has positive density; for base $a = 10$ this density is Artin's constant $C = \prod_q \big(1 - \frac{1}{q(q-1)}\big) \approx 0.3739558$. Hooley [11] proved the conjecture under the Generalized Riemann Hypothesis, Gupta and Murty [8] and Heath-Brown [10] obtained the strongest unconditional results, and Moree's survey [14] gives a comprehensive account. For $a = 10$ the property has a pleasant classical meaning: 10 is a primitive root mod p precisely when the decimal expansion of $1/p$ has maximal period $p - 1$ (the *full reptend* or *long* primes; see [3, pp. 166–171]).

The second theme is the surprising dependence between *consecutive* primes discovered by Lemke Oliver and Soundararajan [12]: for a modulus q, a prime in a given reduced residue class mod q is followed by a prime in the *same* class markedly less often than equidistribution would predict. The phenomenon, which they explain conjecturally via the Hardy–Littlewood prime k-tuple conjectures [9], decays slowly and is prominent at the scales accessible to computation.

This paper asks a question that, to our knowledge, has not been raised before: *are the Artin statuses of consecutive primes correlated?* Since the Artin property of p is governed by the arithmetic of $p - 1$ (which primes q divide it), and since the residues of consecutive primes modulo small q are correlated by [12], one expects *some* induced dependence between $\mathbf{1}[10 \text{ is a primitive root mod } p_n]$ and $\mathbf{1}[10 \text{ is a primitive root mod } p_{n+1}]$. What is not obvious is its sign, its size, its structure as

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a function of the gap $g = p_{n+1} - p_n$, or the extent to which it is connected to known arithmetic structure. We answer these questions computationally for all primes up to 10^9 and isolate one component that is not merely statistical but *deterministic*: an exclusion law for gaps $g \equiv 20 \pmod{40}$ (Theorem 1).

Related but distinct questions have been studied. Garcia, Kahoro, and Luca [6] (see also the sequel [7]) compared the *number* of primitive roots, $\varphi(p-1)$ versus $\varphi(p+1)$, for twin prime pairs, finding a strong bias which they explain under the Bateman–Horn conjecture. Pollack [16] proved (under GRH) that for every nonsquare $g \neq -1$ and every m, there are infinitely many runs of m consecutive primes each having g as a primitive root. Baker and Pollack [2] proved unconditionally that if Q is a set of at least $\exp(Cm)$ multiplicatively independent primes (for an absolute constant C), then infinitely many runs of m consecutive primes each have some element of Q as a primitive root, with the primes lying in a bounded-length interval. We emphasise that these are *existence* results, produced by sieve constructions that select rare favourable configurations; they make no statement about the typical joint behaviour of Artin status along the sequence of primes. The present paper asks the complementary *statistical* question — how the Artin statuses of consecutive primes are correlated on average — which appears to be unexamined; and the exclusion law of Theorem 1, though elementary, seems to be unrecorded.

Recent work has also sharpened the analytic understanding of Artin densities for *individual* primes: Fan and Pollack [5] establish (under GRH) a version of the Artin–Hooley asymptotic that is uniform in the base, and Perucca and Shparlinski [15] prove (under GRH) uniform upper and lower bounds for the Artin density relative to the corresponding Artin constant. These results concern the distribution of Artin primes one prime at a time; to our knowledge the *joint* behaviour of the Artin property at consecutive primes, which is the subject of this paper, has not been considered.

Summary of results.

(1) (Theorem 1) If p and $p+g$ are primes with $p, p+g > 5$ and $g \equiv 20 \pmod{40}$, then $\left(\frac{10}{p+g}\right) = -\left(\frac{10}{p}\right)$; consequently 10 is a quadratic residue for exactly one of the two, and at most one can be an Artin prime for base 10. If instead $g \equiv 0 \pmod{40}$, then $\left(\frac{10}{p+g}\right) = \left(\frac{10}{p}\right)$.

(2) (Section 4) Across all 50,847,530 consecutive prime pairs with $7 \le p_n < p_{n+1} \le 10^9$, Artin statuses are anticorrelated: $\delta = P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) - P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n) = -0.01414$, with Pearson $\chi^2 = 10{,}167$ on one degree of freedom.

(3) (Section 5) The dependence is violently non-uniform in the gap: $\delta(g)$ ranges from -0.592 (at $g = 60$, where the theorem forces exclusion) to $+0.643$ (at $g = 40$), with sign and magnitude predicted by $g \bmod 40$ and $g \bmod 3$.

(4) (Section 6) Conditioning on the pair of residues of (p_n, p_{n+1}) modulo 120 or 840 substantially reduces a selected-cell signed residual metric; the remaining small residual is consistent with further small-modulus channels but has not been shown to vanish. See Section 6 for exact statistics and their limitations.

(5) (Section 7) As an intermediate link we measure $r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big) = -0.0410$: consecutive primes' $p-1$ factorisations repel.

Throughout, *Artin prime* means Artin prime for base 10 unless stated otherwise, and p_n denotes the n-th prime.

2. The exclusion law

Definition 1. For an odd prime $p \nmid 10$, write $\left(\frac{10}{p}\right)$ for the Legendre symbol. We say p is an *Artin prime* (for base 10) if the multiplicative order of 10 modulo p equals $p - 1$.

If 10 is a quadratic residue mod p then its order divides $(p-1)/2$, so a QR is never a primitive root; $\left(\frac{10}{p}\right) = -1$ is a necessary (not sufficient) condition for the Artin property.

Theorem 1 (Exclusion law for gaps $\equiv 20 \pmod{40}$)**.** *Let p and $p' = p+g$ be primes with $p, p' > 5$ and $g > 0$, $g \equiv 20 \pmod{40}$. Then*

$$\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right).$$

In particular, 10 is a quadratic residue modulo exactly one of p, p', and at most one of them is an Artin prime for base 10.

Proof. By multiplicativity, $\left(\frac{10}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{5}{p}\right)$.

Since $g \equiv 0 \pmod 5$, we have $p' \equiv p \pmod 5$. By quadratic reciprocity ($5 \equiv 1 \bmod 4$), $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$ depends only on $p \bmod 5$; hence $\left(\frac{5}{p'}\right) = \left(\frac{5}{p}\right)$.

Since $g \equiv 4 \pmod 8$, we have $p' \equiv p + 4 \pmod 8$. The map $r \mapsto r + 4$ on odd residues mod 8 exchanges $1 \leftrightarrow 5$ and $3 \leftrightarrow 7$. By the second supplementary law, $\left(\frac{2}{p}\right) = +1$ iff $p \equiv \pm 1 \pmod 8$, and the exchange maps $\{1, 7\}$ onto $\{5, 3\}$ and vice versa; hence $\left(\frac{2}{p'}\right) = -\left(\frac{2}{p}\right)$ for every admissible residue.

Multiplying, $\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right)$. Whichever of the two primes has $\left(\frac{10}{\cdot}\right) = +1$ satisfies $\mathrm{ord}(10) \mid (p-1)/2 < p-1$, so it is not an Artin prime. $\square$

Corollary 1 (Coupling law for gaps $\equiv 0 \pmod{40}$)**.** *If $p, p' = p + g$ are primes with $p, p' > 5$ and $g \equiv 0 \pmod{40}$, then $\left(\frac{10}{p'}\right) = \left(\frac{10}{p}\right)$. In particular, if either is an Artin prime, then 10 is a quadratic non-residue for both.*

Proof. Now $g \equiv 0 \pmod 8$ and $g \equiv 0 \pmod 5$, so both $\left(\frac{2}{\cdot}\right)$ and $\left(\frac{5}{\cdot}\right)$ are preserved. $\square$

Remark 1. *The modulus 40 is specific to base 10: it is the conductor of the quadratic character $p \mapsto \left(\frac{10}{p}\right)$, namely $4 \cdot 10 = 40$ (the discriminant of $\mathbb{Q}(\sqrt{10})$). For a general non-square base a the same argument applies with 40 replaced by the conductor $|d_a|$ of the quadratic character attached to $\mathbb{Q}(\sqrt{a})$: gaps $g \equiv 0 \pmod{|d_a|}$ preserve quadratic-residue status. Some bases also admit residue classes of gaps that reverse the Legendre symbol on every admissible pair; their existence requires a separate classification. Base 10 is distinguished only by our decimal habit and the full-reptend interpretation.*

Remark 2. *For gaps in classes other than $0, 20 \pmod{40}$ the relation between $\left(\frac{10}{p}\right)$ and $\left(\frac{10}{p'}\right)$ depends on p as well as g (the shift on $p \bmod 8$ or $p \bmod 5$ is not a symbol-preserving or symbol-flipping involution), producing the partial correlations catalogued in Section 5.*

We verified Theorem 1 computationally as a guard against error: for all 23,956 consecutive prime pairs below 10^7 with gap $\equiv 20 \pmod{40}$, the Legendre symbols flip in every single case, and a direct order computation on a systematic subsample found no doubly-Artin pair. In the full dataset up to 10^9 (Section 5), the 2,195,882 consecutive pairs with $g \in \{20, 60\}$ contain *no* pair in which both primes are Artin — an empirical count of zero, exactly as the theorem requires.

3. Data and computation

For every prime $7 \le p \le 10^9$ we compute: the gap to the neighbouring primes, the number $\omega(p-1)$ of distinct prime factors of $p-1$ (by trial division), the Artin indicator (computed exactly via $10^{(p-1)/q} \not\equiv 1 \pmod p$ for each prime $q \mid p-1$, using GMP modular exponentiation), and small residues of p. A segmented sieve of Eratosthenes generates the primes; the corrected sieve emits all 50,847,531 primes from 7 to 999,999,937 (the largest prime $\leq 10^9$).

Endpoint correction. An earlier version of the sieve contained two endpoint bugs: it emitted a duplicate row for $p = 7$ and omitted the final prime $p = 999{,}999{,}937$. These have been corrected. The duplicate was already skipped by the analysis scripts; the missing endpoint adds one pair at gap 8 (both primes non-Artin: $\mathrm{ord}_{10}(999{,}999{,}937) = 333{,}333{,}312$; independently verified via `sympy`). This correction changes the pair count from 50,847,529 to 50,847,530 and does not materially alter any displayed summary statistic.

As a global check, the number of Artin primes in the dataset is 19,016,617, a proportion 0.373993, matching Artin's constant 0.373956 to four decimal places (for base 10 the squarefree part is $10 \not\equiv 1 \bmod 4$, so no correction factor applies and the conjectural density is exactly C; see [11, 14]). Agreement with a conjectural density is a sanity check, not a verification of every indicator.

The consecutive-pair analysis is a single streaming pass over the 50,847,530 ordered pairs (p_n, p_{n+1}), accumulating 2×2 contingency tables of $(\mathbf{1}[\mathrm{Art}_n], \mathbf{1}[\mathrm{Art}_{n+1}])$: globally, stratified by gap g, and stratified by joint residues of (p_n, p_{n+1}) to various moduli. All counts reported below are exact (no sampling), except where noted. Analysis scripts and their outputs are available with the data; see the Data Availability statement.

Runtime, hardware, and validation. All computations were performed on a single machine with a 2-core Intel Xeon E5-2673 v4 processor at 2.3 GHz and 8 GB of RAM, running Ubuntu Linux; no parallelism beyond a single core was used for any run reported here. The segmented sieve (segment length 2^{21}, so the working set stays within cache) together with the per-prime computation — trial-division factorisation of $p - 1$ and the Artin test by modular exponentiation — generates the full dataset to 10^9; peak memory usage is under 200 MB, as only one segment and the small primes to $\sqrt{x}$ are held in memory. The streaming pair analysis reads the dataset once and runs in minutes. As validation we checked: (i) the prime count $\pi(10^9) = 50{,}847{,}534$ against the known value; (ii) the Artin prime count 19,016,617 (proportion 0.373993) against Artin's constant 0.373956, as discussed above; (iii) Theorem 1 on all 23,956 pairs with gap $\equiv 20 \pmod{40}$ below 10^7, with an independent `sympy` order computation on a subsample; and (iv) the mod-12 consecutive-residue transition matrix against the values reported in [12] at comparable scale.

4. The global anticorrelation

Table 1. Artin status of consecutive primes, all 50,847,530 pairs with $7 \leq p_n < p_{n+1} \leq 10^9$.

Quantity	Value
$P(\mathrm{Art}_{n+1})$	0.373993
$P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n)$	0.365141
$P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n)$	0.379281
δ = difference	-0.014140
ϕ coefficient	-0.014140
χ^2 (1 df)	10,167
z-score (nominal)	-100.8

Table 1 presents the central fact: an Artin prime is followed by another Artin prime about 1.4 percentage points less often than a non-Artin prime is. The Pearson statistic is 10,167 (signed square root -100.8 under the nominal independent-status reference model). This is a deterministic finite census, not a random survey; the statistic is reported as a descriptive measure of the strength of association, and should not be interpreted as a sampling confidence level without specifying and justifying a null stochastic model. Consecutive pairs overlap, and residue-cell memberships are serially structured.

The sign is what the Lemke Oliver–Soundararajan repulsion predicts. Heuristically: the Artin property requires $\left(\frac{10}{p}\right) = -1$, which is a condition on $p \bmod 40$; it is further favoured by $3 \nmid p - 1$, i.e. $p \equiv 2 \pmod 3$, since a factor of 3 in $p - 1$ contributes the largest single suppression $(1 - \frac{1}{3})$ of the primitive-root density among odd primes. Consecutive primes repel in their residues modulo 40 and modulo 3 [12]; conditions favourable to the Artin property therefore tend *not* to repeat at p_{n+1} when they hold at p_n, depressing $P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n)$.

For reference, the mod-12 transition matrix in our data displays the repulsion prominently: the diagonal entries $P(p_{n+1} \equiv r \mid p_n \equiv r)$ are 0.1813–0.1815 for $r \in \{1, 5, 7, 11\}$, against the equidistribution value 0.25 — a 27% deficit, consistent in magnitude with the computations in [12] at comparable scales.

5. Gap structure

Stratifying by the gap $g = p_{n+1} - p_n$ reveals sign-alternating within-gap effects that are individually far larger in magnitude than the global $\delta = -0.0141$. The global value is *not* the pair-frequency-weighted average of the within-gap values: for a marginal conditional risk difference such as δ, the strata are additionally reweighted by the gap-conditional distribution of Art_n, so no such averaging identity holds. Indeed the pair-weighted mean of the within-gap differences over the 36 qualifying gap classes (50,114,759 pairs, 98.6% of the census) is +0.00159, opposite in sign to the global value. The two quantities answer different questions, and the gap table below should be read as a decomposition of *where* the association lives, not as a summand decomposition of δ itself (cf. the caution in Section 6). Table 2 shows all gaps with at least 10^5 pairs up to $g = 60$; additional qualifying gaps ($g \in \{62, 64, 66, 70, 72, 78\}$) are included in the analysis scripts and archived results but omitted from the table for brevity. Figure 1 displays $\delta(g)$ with the residue class of $g \bmod 40$ marked, including all qualifying gaps.

Table 2. Conditional Artin dependence by gap ($N \geq 10^5$ pairs, $g \leq 60$). $\delta(g) = P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) - P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n)$.

g	N	$P(A\|A)$	$P(A\|\neg A)$	$\delta(g)$	g	N	$P(A\|A)$	$P(A\|\neg A)$	$\delta(g)$
2	3,424,504	0.291	0.293	−0.002	32	683,896	0.176	0.386	−0.209
4	3,424,679	0.562	0.381	+0.181	34	719,531	0.445	0.445	+0.001
6	6,089,791	0.407	0.382	+0.025	36	1,171,524	0.559	0.314	+0.245
8	2,695,109	0.160	0.409	−0.249	38	548,746	0.288	0.293	−0.006
10	3,484,767	0.448	0.443	+0.005	40	648,780	0.899	0.257	+0.643
12	4,468,957	0.512	0.283	+0.229	42	954,456	0.380	0.358	+0.023
14	2,464,565	0.298	0.296	+0.002	44	389,634	0.344	0.249	+0.095
16	1,846,097	0.296	0.543	−0.247	46	334,720	0.469	0.469	+0.000
18	3,351,032	0.378	0.359	+0.019	48	577,247	0.216	0.454	−0.238
20	1,824,043	**0.000**	0.553	−0.553	50	328,066	0.311	0.307	+0.005
22	1,569,679	0.444	0.447	−0.004	52	245,804	0.572	0.381	+0.191
24	2,367,474	0.295	0.411	−0.116	54	410,754	0.382	0.356	+0.026
26	1,119,585	0.313	0.313	+0.000	56	211,462	0.225	0.389	−0.164
28	1,219,243	0.619	0.366	+0.253	58	181,948	0.437	0.441	−0.004
30	2,177,991	0.390	0.363	+0.027	60	371,839	**0.000**	0.592	−0.592

Three features stand out.

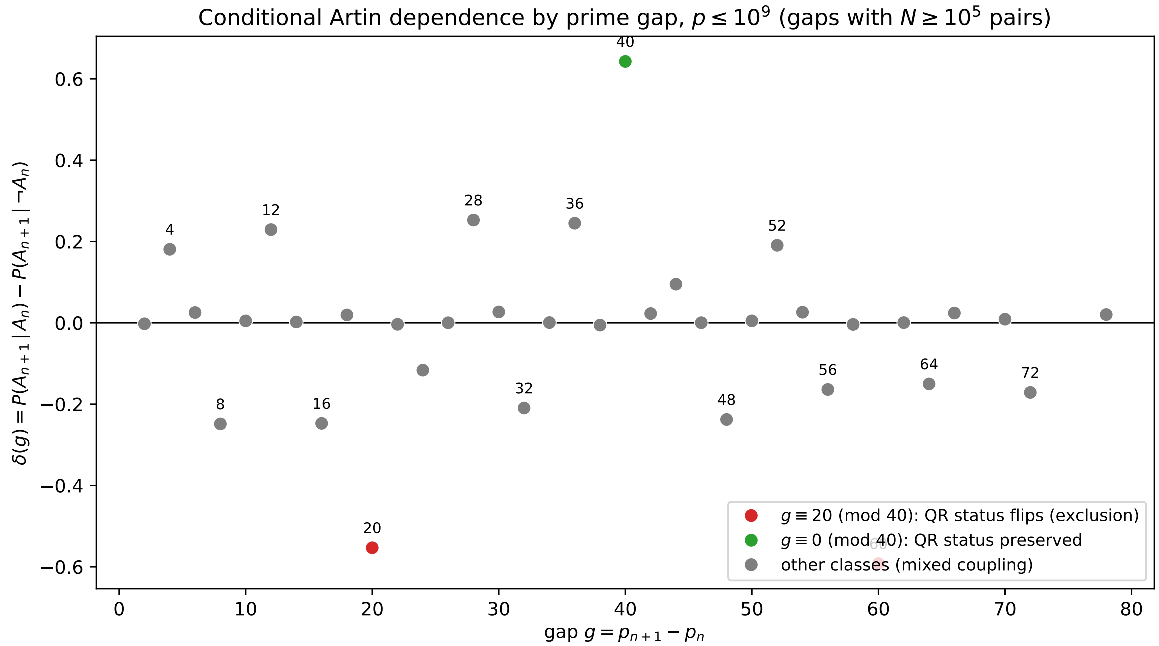

FIGURE 1. $\delta(g)$ for all gaps with $N \ge 10^5$ pairs, $p \le 10^9$. Red: $g \equiv 20 \pmod{40}$ (Theorem 1 forces $P(A|A) = 0$). Green: $g \equiv 0 \pmod{40}$ (Corollary 1 preserves quadratic-residue status). Grey: other classes.

(i) The exclusion rows. At $g = 20$ (1,824,043 pairs) and $g = 60$ (371,839 pairs), $P(A \mid A)$ is exactly zero: among 2.2 million consecutive pairs, not one is doubly Artin. This is Theorem 1 in action, and conversely the observation that motivated its proof.

(ii) The $g = 40$ spike. $P(A \mid A) = 0.899$, the largest conditional probability in the table and 2.4 times the unconditional density. Two forced mechanisms combine. First, by Corollary 1, if p_n is Artin then 10 is a non-residue for p_{n+1} as well, a necessary condition for the Artin property. Second, $40 \equiv 1 \pmod 3$ forces the pair's mod-3 residues: $p_n \equiv 2 \pmod 3$ would give $3 \mid p_n + 40$, impossible; so every $g = 40$ pair has $p_n \equiv 1$, $p_{n+1} \equiv 2 \pmod 3$ — that is, $3 \mid p_n - 1$ (suppressing Art_n) and $3 \nmid p_{n+1} - 1$ (favouring Art_{n+1}). Conditioning on Art_n therefore selects pairs in which p_{n+1} enjoys both the non-residue status and the mod-3 advantage. These conditions are consistent with the large observed value 0.899 but do not quantitatively derive it without an explicit density calculation.

(iii) Sign classes. Gaps divisible by 3 but not constrained mod 40 (e.g. $6, 18, 30, 42, 54$) show small positive $\delta \approx +0.02$: the pair shares its mod-3 status, weakly coupling the largest single suppression factor. Gaps $g \equiv 8, 16 \pmod{24}$-type classes with adverse mod-8/mod-5 shifts show strong negative δ. The sign pattern of Table 2 is reproduced by tracking how the shift by g permutes the residues of p modulo 8, 5, and 3.

6. Residue conditioning

How much of the global association do the explicit residue channels account for? We condition on the *joint* residue class of (p_n, p_{n+1}) to modulus m and measure the residual within-cell dependence. The statistic reported is the sum of per-cell Pearson χ^2 values over all non-degenerate cells meeting a minimum count threshold, together with the pair-weighted mean of per-cell δ. This is a descriptive aggregate of per-cell associations; it is *not* the standard Cochran–Mantel–Haenszel statistic (which would be a single one-degree-of-freedom test of a common association).

Conditioning on $m = 120 = \mathrm{lcm}(8, 3, 5)$ fixes the quadratic character $\left(\frac{10}{\cdot}\right)$ of both primes (its conductor 40 divides 120) and the mod-3 channel. Cells in which 10 is a quadratic residue for p_n or p_{n+1} are degenerate for the analysis (one margin of the 2×2 table vanishes — the Artin property is impossible); the informative population is roughly a quarter of all pairs, those in doubly-non-residue cells. The selected populations in Table 3 are the thresholded subsets of this population: 12,255,203 pairs (24.10% of the census) at modulus 120 and 12,086,765 pairs (23.77%) at modulus 840. The unthresholded doubly-non-residue fraction need not be exactly 1/4.

Table 3. Selected-cell residual association after conditioning on joint residues mod m. "δ_{res}" is the pair-weighted mean within-cell δ over non-degenerate cells meeting the count threshold; "cells" is the number of such cells; "pairs used" is the total pairs in those cells. Cells with fewer than 5000 ($m = 120$) or 2000 ($m = 840$) pairs are excluded.

Conditioning	cells	pairs used	$\sum \chi^2$ (df)	δ_{res}
none (global)	1	50,847,530	10,167 (1)	−0.014140
mod 120 joint residues	145	12,255,203	789.2 (145)	−0.001055
mod 840 joint residues	633	12,086,765	719.8 (633)	−0.000473

Interpretation and limitations. The selected-cell signed mean residual δ_{res} shrinks from −0.0141 (global) to −0.00106 (mod 120, 145 cells covering 12.3 million pairs) to −0.00047 (mod 840, 633 cells covering 12.1 million pairs). The summed per-cell χ^2 at mod 840 is 719.8 on 633 nominal degrees of freedom ($\chi^2/\mathrm{df} = 1.14$). Two consistency observations: (i) the residual shrinks monotonically as channels are absorbed ($-0.0141 \to -0.00106 \to -0.00047$); (ii) a split-sample check (pairs with $p < 5 \times 10^8$ versus $p \geq 5 \times 10^8$, conditioning mod 120) gives residuals −0.00096 and −0.00094 — stable across the range.

Several caveats apply. The global δ over 50.8 million pairs and the selected-cell δ_{res} over 12.1 million selected informative pairs are *different estimands on different populations*; arithmetic comparison of their magnitudes is descriptive but does not constitute a proper decomposition. A marginal risk difference is not, in general, the pair-weighted mean of stratum risk differences without a between-group term and specified weights. The per-cell Pearson calibration assumes independent observations within each cell, which is not established for this serially structured deterministic census. The paper does not specify a numerical model for the omitted channels ($q \geq 11$) against which "compatible" could be formally tested.

Joint-residue conditioning substantially reduces a selected-cell signed residual association. The remaining association is small in that metric but is not established to vanish. These results are consistent with a substantial role for arithmetic residue structure; they do not constitute a complete decomposition or an independent quantitative verification of the Hardy–Littlewood-based explanation of the consecutive-prime bias.

7. The intermediate link: repulsion of $\omega(p-1)$

The chain from residue bias to Artin status passes through the factorisation of $p - 1$. Its first summary statistic is $\omega(p-1)$, the number of distinct prime factors, whose average grows like $\log\log p$ (for $p-1$ shifted primes this is a classical result of Erdős [4]). If consecutive primes repel modulo every small q, then $q \mid p_n - 1$ makes $q \mid p_{n+1} - 1$ *less* likely than average, and summing over q one predicts a negative correlation between $\omega(p_n - 1)$ and $\omega(p_{n+1} - 1)$.

The data confirm this with a value that is, to our knowledge, unreported:

$$r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big) = -0.0410 \qquad (50{,}847{,}530 \text{ pairs}, \ p \leq 10^9).$$

The factorisations of $p-1$ for consecutive primes are not independent: they repel, in the same sense and for the same reason that the primes themselves repel in residue classes. The Pearson coefficient is reported as a descriptive nominal value.

8. Open questions

The Hardy–Littlewood-based framework of [12] makes the residue transition frequencies of consecutive primes quantitatively predictable, and every channel in our conditioning is built from such frequencies. This motivates the following open questions.

Conjecture 1 (Persistence)**.** *For primes up to x, the consecutive-pair Artin correlation $\delta(x) = P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) - P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n)$ is negative for all sufficiently large x.*

The measurements $\delta(10^8) = -0.01336$ and $\delta(10^9) = -0.01414$ do not determine whether $\delta(x) \to 0$ as $x \to \infty$ or tends to a small nonzero limit, and we do not claim a prediction either way. We note one reason for caution against assuming a nonzero limit, in a restricted setting. Consider a *separable* fixed-modulus channel model: write R, S for the residue classes of (p_n, p_{n+1}) modulo a fixed M, and suppose the modelled indicators satisfy $\mathbb{E}[X \mid R, S] = \alpha(R)$, $\mathbb{E}[Y \mid R, S] = \beta(S)$ and $\mathrm{Cov}(X, Y \mid R, S) = 0$, so that the only modelled dependence between the two endpoints runs through the joint law of (R, S). If that joint law tends to the product of its marginals, as the leading term of the Lemke Oliver–Soundararajan prediction gives for fixed M [12], then $\mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] \to 0$, and hence the modelled $\delta \to 0$ whenever the limiting source marginal is nondegenerate. Non-uniform singular-series weights across gap classes do not by themselves defeat this conclusion: the gaps grow with x, and it is the surviving total covariance, rather than the non-uniformity of the weights, that would have to be estimated.

The separability hypothesis is essential and is not automatic. Conditioning on the joint pair (R, S) alone does not force independence: taking R, S independent and uniform on the sixteen reduced classes modulo 40 and setting $X = Y = Q(R)Q(S)$, where Q indicates the eight classes on which the relevant quadratic symbol is -1, gives perfectly equidistributed residue pairs together with $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[XY] = \frac{1}{4}$ and $\mathrm{Cov}(X, Y) = \frac{3}{16}$. So the observation above constrains separable finite-modulus models, not every model built on a fixed modulus. Extending a finite-obstruction approach to a statement about the limit would in addition require joint-distribution information at each finite channel together with a uniform bound on the omitted higher-power-residue obstructions; the quadratic channel visible here is only the first of these. The question of the limiting behaviour is unresolved; finite-range measurements cannot distinguish the two alternatives.

Research question (residue completeness). Does there exist, for every $\varepsilon > 0$, a modulus M such that conditioning on the joint residues of (p_n, p_{n+1}) modulo M reduces some appropriate within-cell dependence measure below ε in absolute value? This is stated as a question rather than a conjecture because a precise formulation requires specifying the order of limits (over x and M), the treatment of sparse and degenerate cells, the dependence criterion (e.g. a covariance decomposition rather than a signed cell mean), and the range of x over which the bound is asserted. We do not know a formulation that avoids these ambiguities.

We remark that a sufficiently uniform version of the Hardy–Littlewood prime pair conjectures would predict the joint distribution of $(p_n \bmod M, p_{n+1} \bmod M, g)$ via singular series; whether such predictions carry the Chebotarev/power-residue information needed for Artin status, and whether the no-intermediate-prime condition defining consecutiveness introduces additional structure, requires further analysis.

Conjecture 2 (ω repulsion)**.** *$r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big)$ is negative for all large x and tends to 0 as $x \to \infty$.*

The single reported value does not establish eventual sign or limit. We state this as a conjecture consistent with the data.

9. Discussion

What is new here. The individual ingredients are classical: quadratic reciprocity, the quadratic-residue obstruction to primitive roots, and the consecutive-prime bias of [12]. The contributions of this paper are (i) the observation that these combine into a deterministic exclusion law for prime pairs at gaps $\equiv 20 \pmod{40}$ (Theorem 1) — elementary, but we have not found it stated elsewhere; (ii) a measurement of the consecutive-prime Artin correlation, its gap profile, and the ω repulsion; and (iii) evidence that arithmetic residue structure is substantially connected to the measured dependence. The theorem is an immediate application of standard quadratic reciprocity; its simplicity is a feature, not a limitation.

Relation to runs of Artin primes. Pollack [16] proved (under GRH) that for every nonsquare $g \neq -1$ and every m, there are infinitely many runs of m consecutive primes each having g as a primitive root. Baker and Pollack [2] proved unconditionally that if Q contains at least $\exp(Cm)$ multiplicatively independent primes, then infinitely many runs of m consecutive primes each have some element of Q as a primitive root, with the primes lying in a bounded-length interval. Our results add an arithmetic constraint to the base-specific version: for base 10, no run of all-Artin consecutive primes can cross a gap $\equiv 20 \pmod{40}$. We have not inferred longer-run frequencies from the pair statistic.

Future work. Three extensions suggest themselves. (1) Other bases: the same pipeline with $a = 2, 3, 5, 6, 7$ would test the conductor prediction of the Remark after Corollary 1 (e.g. conductor 8 for $a = 2$, so exclusion classes mod 8; conductor 24 for $a = 6$). (2) Scale: extending to 10^{12} would help distinguish decay from persistence. (3) Theory: a derivation of $\delta(g)$ from the singular series of [9] combined with the density computations of Hooley's method, at least for the dominant channels, appears feasible and would replace our empirical conditioning with a formula.

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Declaration of interest

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Data availability

All code required to reproduce every number in this paper is supplied in the accompanying reproduction package (C segmented sieve, Python analysis scripts, figure script, and raw output logs). The corrected code, manuscript and summary outputs are at `repositoryURLwithheldforreview`; the full per-prime CSV is not included in the public repository. The sieve limit is configurable via a command-line argument; within the Paper-1 reproduction pipeline described above, the scripts accept their input/output paths as arguments and default to repository-relative locations. This statement is scoped to that pipeline and does not extend to other scripts that may be present in the repository for related projects. The per-prime dataset itself (1.8 GB CSV covering all 50,847,531 primes from 7 to 999,999,937: gap, $\omega(p-1)$, Artin indicator, and small residues for each prime) is not deposited because of its size, but it is exactly and deterministically reproducible from the included C source on a single machine; it is also available from the author on request.

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